

ELECTIVE-IV

UEC751: DSP PROCESSORS

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Course Objective: To enhance comprehension capabilities of students through fundamentals of DSP processors with its architecture, memory architecture along with instruction set, interrupts, pipelining and different types of DSP processors.				
Syllabus				
An Introduction to DSP Processors: Advantages of DSP, characteristics of DSP systems, classes of DSP applications, DSP processor embodiment and alternatives, Fixed and floating point number representation, IEEE 754 format representation, Fixed Vs Floating point processors.				
DSP Architecture: An introduction to Harvard Architecture, Differentiation between Von-Neumann and Harvard Architecture, Quantization and finite word length effects, Bus Structure, Central Processing Unit, ALU, Accumulators, Barrel Shifters, MAC unit, compare, select, and store unit (CSSU), data addressing and program memory addressing.				
Memory Architecture: Memory structures, features for reducing memory access required, wait states, external memory interfaces, memory mapping, data memory, program memory and I/O memory, memory mapped registers.				
Addressing and Instruction Set: Various addressing modes - implied addressing, immediate data addressing, memory direct addressing, register direct and indirect addressing, and short addressing modes, Instruction types, various types registers, orthogonality, assembly language and application development.				
Interrupts and Pipelining: Interrupts, pipelining and performance, pipelining depth, interlocking, interrupt effects, instruction pipelining.				
Processors: Architecture and instruction set of TMS320C3X, TMS320C5X, Status register 0 (ST0), Status register 1 (ST1), Circular buffer control register (CBCR), Processor mode status register (PMST), some example programs, An introduction to TMS320C6713 and TMS320C5517 Kit, <i>IEEE 754, ANSI standard TMS320</i>				
Laboratory Work: NA				
Course Learning Objectives (CLO)				
The students will be able to:				
1. Differentiate between generalised processor and DSP processor.				
2. Analyze special characteristics and features of generalized DSP processors.				
3. Understand the concept of memory architectures and addressing modes for generalized DSP processors.				
4. Understand pipelining and programming concept of TMS320C3X and TMS320C5X.				
5. Understand detailed architectures, different registers and instruction sets of TMS320C3X and TMS320C5X.				
Text Books				
1. Lapslev, P., Bier, J., Shoham, A. and Lee, E.A., <i>DSP Processor Fundamentals:</i>				

- Architecture and Features, IEEE Press Series on Signal Processing, IEEE (2000).*
2. Venkataramani, B. and Bhaskar, M., *Digital Signal Processor: Architecture, Programming and Applications*, Tata McGraw Hill (2003).
3. TI DSP reference set (www.ti.com).

Reference Books

1. Padmanabhan, K., Ananthi, S. and Vijayarajeswaran, R., *A practical Approach to Digital Signal Processing*, New Age International Pvt. Ltd (2001).
2. Babast, J., *Digital Signal Processing Applications using the ADSP-2100 family*, PHI (1992).